

Job Sequencing and Tool Switching

Structural results and a branch-and-Benders-cut study

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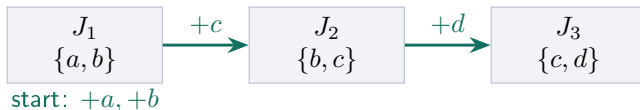
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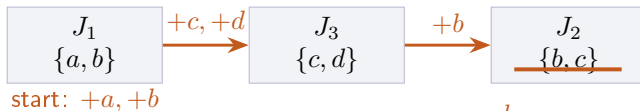
Magazine capacity $b = 2$; the machine starts empty.

order 1, 2, 3



4 insertions

order 1, 3, 2



5 insertions

b was removed, then returned

Loading is easy; the order is the hard part

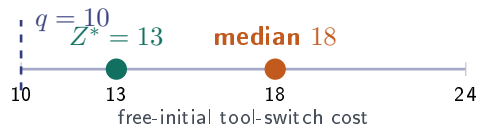
Fixed order

When a tool must leave, remove the loaded tool whose next use is furthest away.

$\text{KTNS}(\pi)$ in $O(n|U|)$.

Choosing the order is NP-hard for every fixed $b \geq 2$.

40,320
orders of the eight-job instance L1-1
180
are optimal



(Tang & Denardo, 1988; Crama et al., 1994)

CNC cells
tool magazine

PCB assembly
feeder slots

Colour printing
ink cartridges

Order picking
what a picker carries

Same combinatorial core

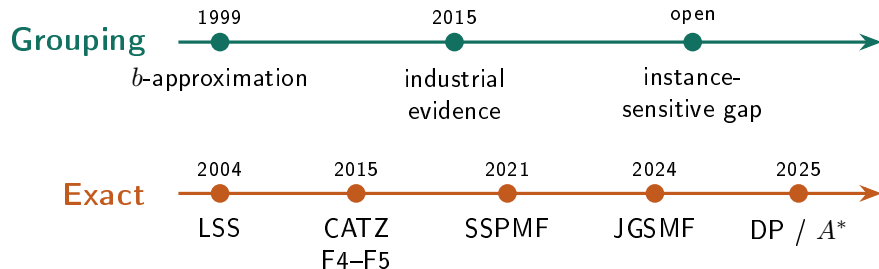
A capacity- b resource, jobs that need subsets of it, and a changeover cost paid between consecutive jobs.

Why the order matters

A cell running thousands of parts a day repeats the changeover constantly, so the sequence is chosen once and paid for every shift.

(Burger et al., 2015)

Two gaps the literature leaves open



Two things the timeline does not contain: a bound on the two-phase loss that depends on the **instance**, and a relaxation that gets **past the counting bound**.

(Crama & van de Klundert, 1999; Catanzaro et al., 2015; da Silva et al., 2021)

Two questions, and how the talk is organised

Question 1 – structural

Industry does not solve the SSP. It groups jobs into as few magazine-fitting batches as possible, then sequences the batches.

What does insisting on the fewest groups cost, once the final KTNS step is allowed to re-optimize the loading?

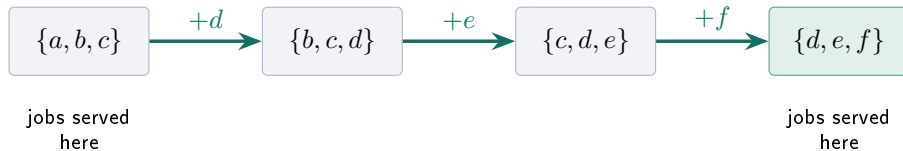
Question 2 – algorithmic

Every compact model surveyed relaxes to the counting bound q , so it is fast where q is exact and adrift where it is not.

Can a method that never forms that relaxation – an exact loading oracle, or a reformulation over configurations – get past it?

Part I answers the first and finds an unbounded loss. **Part II** answers the second, and the answer is no — for a reason worth naming.

The configuration view, and where q comes from



A schedule is a walk

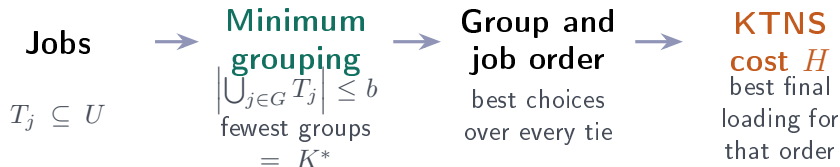
Magazine configurations are the nodes; the step cost is $d(C, C') = |C' \setminus C|$, the tools that must be inserted. Every job must be served at some node that contains its requirement.

Hence one counting bound

Each used tool enters at least once, and at most b of them can already be loaded at the start:

$$Z^* \geq q = |U| - b.$$

This single quantity governs **both** halves of the talk.



$$Z^*$$

best KTNS cost over every job order

$$H - Z^*$$

loss that remains even under favourable tie-breaking

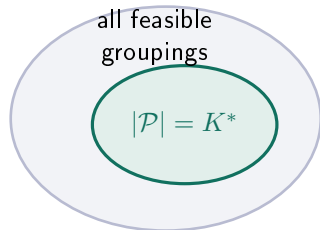
(Crama & van de Klundert, 1999)

Drop the fewest-groups rule: grouping is exact

$$Z^* = \min_{\mathcal{P} \text{ feasible}} \gamma(\mathcal{P})$$

$$H_{\text{walk}} = \min_{\substack{\mathcal{P} \text{ feasible} \\ |\mathcal{P}|=K^*}} \gamma(\mathcal{P}), \quad H \leq H_{\text{walk}}.$$

The approximation enters when the first phase keeps only the groupings with the fewest groups.



$$\text{cost} = q + R$$

$q = |U| - b$; R counts reinsertions.

Four jobs are enough for a genuine grouping loss

Required tools; capacity $b = 4$

	1	2	3	4	5	6	7
A	●	●					
B			●	●			
C	●		●		●	●	
D		●			●	●	●

A and B are the only feasible pair.

Unrestricted order

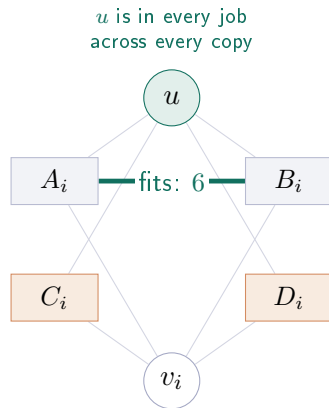
$$B, C, A, D \implies Z^* = q = 3.$$

Every minimum grouping

$$A \text{ and } B \text{ consecutive} \implies H = 4.$$

One forced adjacency creates one reinsertion.

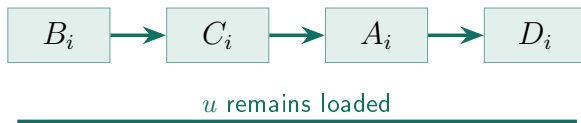
One shared tool makes the family connected



- C_i and D_i fill all six slots.
- $A_i \cup B_i$ uses exactly six tools.
- Small jobs from different copies use seven.

unique minimum
grouping
 $K^* = 3g$

The unrestricted order reaches coverage



repeat copy by copy; every newly loaded private tool is appearing for the first time

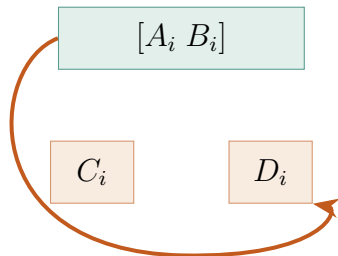
$$|U| = 8g + 1, \quad b = 6$$

$$q = |U| - b = 8g - 5.$$

No reinsertion is needed, so the coverage lower bound is attained:

$$Z^* = q = 8g - 5.$$

Every minimum grouping forces one return per copy



**at least one private tool
must return**

Lower bound

$$H \geq q + g = (8g + 1 - 6) + g = 9g - 5.$$

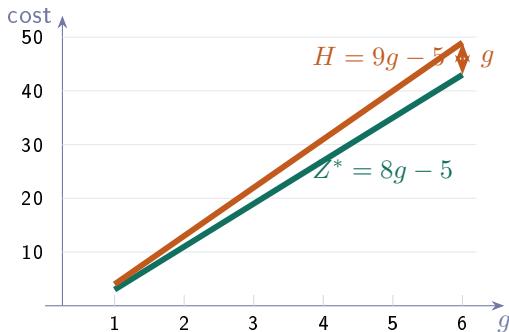
Matching grouped order

$$A_i, B_i, C_i, D_i$$

has exactly one return per copy. Therefore

$$H = 9g - 5.$$

The additive gap grows without bound



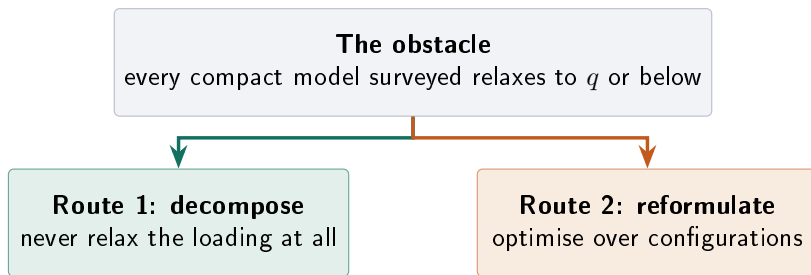
$$H - Z^* = g$$

The additive loss is unbounded, even though all jobs remain connected through u .

$$\frac{H}{Z^*} \rightarrow \frac{9}{8}.$$

This family does not prove a universal $9/8$ ratio bound.

Two routes past the counting bound



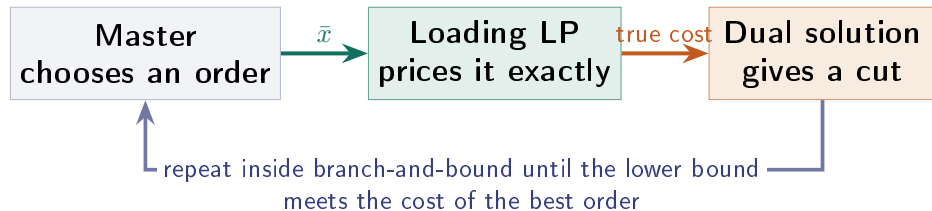
Branch-and-Benders-cut

The loading subproblem is polynomial, so it becomes an exact oracle and the master never forms a weak relaxation *of the loading*.

Position-indexed branch-and-price

Columns are magazine configurations. One variant prices *transitions*, and its relaxation can provably exceed q .

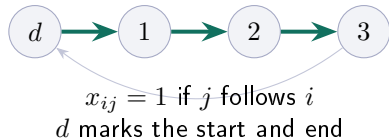
(da Silva et al., 2021; Rahmaniani et al., 2017)



The decomposition is natural because the fixed-order subproblem is polynomial and exact.

(Hooker & Ottosson, 2003; Codato & Fischetti, 2006)

The master starts from local and coverage bounds



$$\begin{aligned} \min \theta, \quad & \sum_{i \neq j} x_{ij} = 1, \quad \sum_{j \neq i} x_{ij} = 1, \\ & \sum_{i,j \in S} x_{ij} \leq |S| - 1 \quad (S \subseteq J). \end{aligned}$$

Starting lower bounds

$$\theta \geq |U|, \quad \theta \geq \sum_{i,j \in J} w_{ij} x_{ij} + \sum_{j \in J} |T_j| x_{dj},$$

where $w_{ij} = \max(0, |T_i \cup T_j| - b)$.

The loading LP is integral on a fixed path

$$\begin{aligned} \min \quad & \sum_{j,t} z_{jt} \\ \text{s.t.} \quad & y_{jt} \geq 1 & (t \in T_j), \\ & \sum_t y_{jt} \leq b & (j \in J), \\ & z_{jt} \geq y_{jt} - y_{it} - (1 - x_{ij}) & (i \neq j), \\ & z_{jt} \geq y_{jt} - (1 - x_{dj}) & (j \in J), \\ & y, z \geq 0. \end{aligned}$$

(Tang & Denardo, 1988)

y_{jt} records whether tool t is loaded at job j .

z_{jt} charges an insertion when the selected path moves into j .

fixed path
↓
exact KTNS cost

The dual cut is supported only on arcs

$$\theta \geq \alpha + \sum_{i \neq j} \beta_{ij} x_{ij}, \quad \beta_{ij} = \sum_t \bar{\lambda}_{ijt}.$$

Valid globally

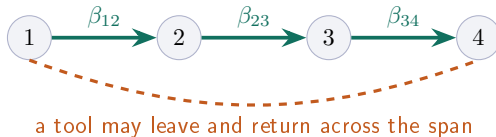
The dual feasible region does not depend on the proposed order.

Tight locally

An optimal dual prices the generating order exactly.

Arc-supported

Every learned coefficient sits on one predecessor arc.



PCF one configuration per position

$$\min \sum_{t,k} w_t^k \quad \text{s.t.} \quad \sum_C y_C^k = 1, \quad \sum_{k \geq 2} w_t^k + \sum_{C \ni t} y_C^1 \geq 1.$$

Without the counting rows the relaxation is 0; with them it is exactly q .

PCF' prefix form

Removes the symmetry of empty positions.
Same relaxation value.

PTF one *transition* per step

$$\min \sum_k \sum_{(C,C')} d(C,C') z_{C,C'}^k$$

A column carries a step, so the relaxation cannot avoid paying for it:

$$\text{LP(PTF)} \geq q, \quad \text{and can exceed it.}$$

The price

Pricing is NP-hard, and PTF prices coupled *pairs*, so a node costs far more. Rows stay $O(n|T|)$ and fixed.

The campaign tests 12 solver configurations

Compact baselines

SSPMF

CATZ-F4

LSS

Reimplementations used for a controlled comparison, not a literature-wide ranking.

Benders regimes

BBC-LP

BBC-LP+T

BBC-K

BBC-LP+F

BBC-K+F

BBC-LP+F+H

BBC-LP+F+C

BBC-LP+F+P

BBC-LP+ACC

1,410 canonical benchmark instances

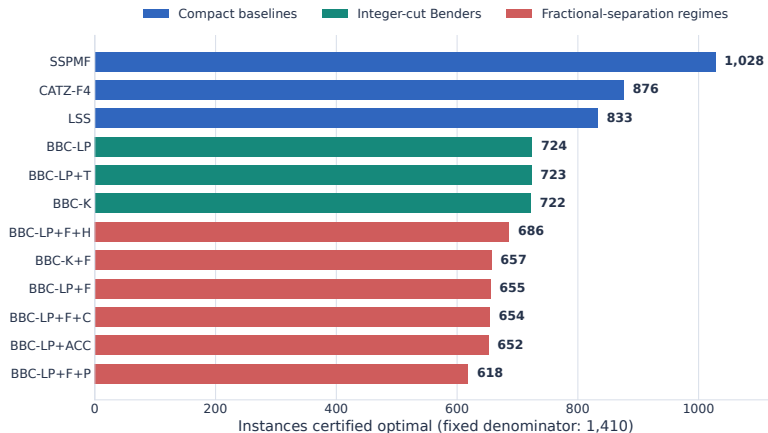
12 solver configurations

16,920 canonical outcomes

1 h per run, fixed denominator

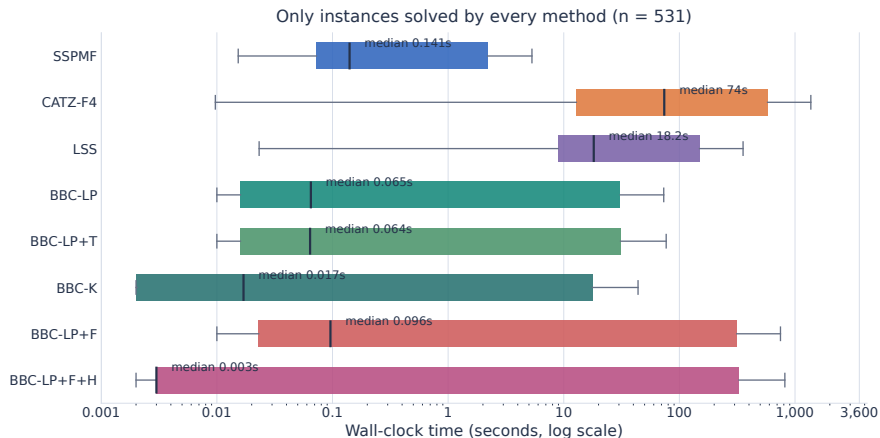
(Laporte et al., 2004; Catanzaro et al., 2015; da Silva et al., 2021)

Every compact baseline certifies more than Benders



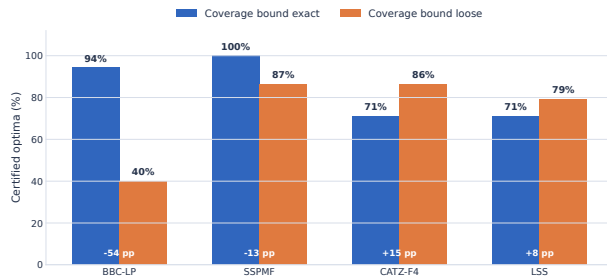
Each bar uses the same 1,410-instance denominator.

Benders is quick, with a long tail

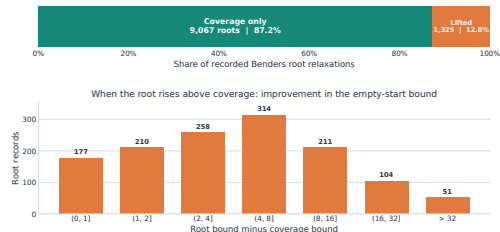


The same 531 instances appear in every box; the horizontal axis is logarithmic.

Closure drops when the bound misses the optimum



1,107 instances with a known optimum.



223/383 time-limited,
known-optimum
BBC-LP incumbents already equal
the optimum.

What the theory isolates

Minimum grouping can force a tool out and back:

$$\text{cost} = q + \text{reinsertions}.$$

What the campaign isolates

The tested master often finds the right order but cannot certify those returns.

The work narrows the next experiment; it claims no new SOTA solver.

Next exact tests

- disaggregate θ by position;
- encode job-union structure over a window;
- test the stronger bound under a matched protocol.

Use the grouping result

- search $K^*, \dots, K^* + \delta$ groupings;
- flatten each with KTNS; use the orders as seeds.

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- Catanzaro, D., Gouveia, L., and Labbé, M. (2015). Improved integer linear programming formulations for the job sequencing and tool switching problem. *European Journal of Operational Research*, vol. 244, no. 3, pp. 766–777.
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- Hooker, J. N., and Ottosson, G. (2003). Logic-based Benders decomposition. *Mathematical Programming*, vol. 96, pp. 33–60.
- Laporte, G., Salazar-González, J. J., and Semet, F. (2004). Exact algorithms for the job sequencing and tool switching problem. *IIE Transactions*, vol. 36, no. 1, pp. 37–45.
- Rahmaniani, R., Crainic, T. G., Gendreau, M., and Rei, W. (2017). The Benders decomposition algorithm: a literature review. *European Journal of Operational Research*, vol. 259, no. 3, pp. 801–817.
- Tang, C. S., and Denardo, E. V. (1988). Models arising from a flexible manufacturing machine, Part I: minimization of the number of tool switches. *Operations Research*, vol. 36, no. 5, pp. 767–777.

Every work listed here is cited on a slide of this talk. The full bibliography is in the report.

Thank You!

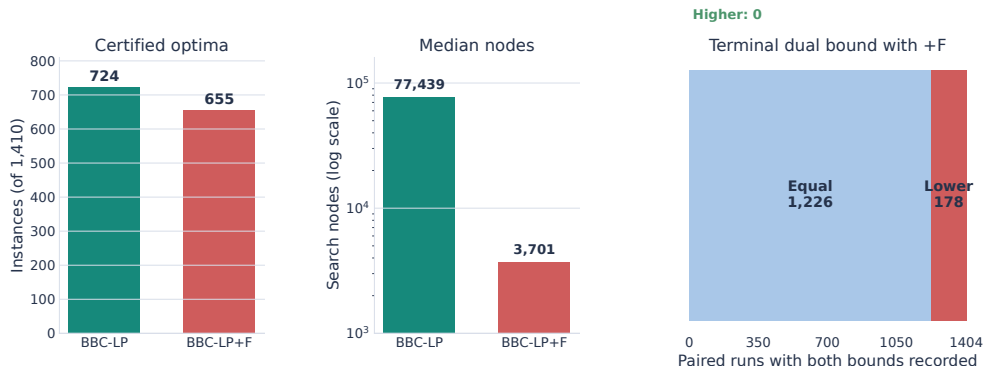
Questions

Backup: certified optima by method and family

Configuration	Cat.	Crama	Lap.3	Lap.4	Lap.5	Lap.7	All
SSPMF	62	68	340	330	162	66	1028
CATZ-F4	44	44	340	330	74	44	876
LSS	41	42	340	330	39	41	833
BBC-LP+T	34	34	340	192	84	39	723
BBC-LP	34	35	340	192	84	39	724
BBC-K	35	35	340	190	83	39	722
BBC-LP+F+H	39	37	316	157	97	40	686
BBC-LP+F	33	32	313	156	85	36	655
BBC-LP+F+C	33	32	311	157	85	36	654
BBC-LP+ACC	39	37	286	154	97	39	652
BBC-K+F	33	31	316	156	86	35	657
BBC-LP+F+P	31	32	282	155	83	35	618

Family denominators: 160, 160, 340, 330, 340, 80; total 1,410.

Backup: fractional cuts spend more, certify less



BBC-LP+F generated 67.0 million fractional cuts. The +F regime also disables presolve.

Against BBC-LP+F: $+H$ certifies 31 more; $+C$ certifies 1 fewer; $+P$ certifies 37 fewer; all three together certify 3 fewer.

Connected-family proof

$K^* = 3g$	unique feasible pair in each copy
$Z^* = 8g - 5$	coverage bound plus a schedule with no return
$H = 9g - 5$	one forced return per copy plus a matching order
connected	universal tool u lies in every job

Independent checks

349,872	loading checks against an exact dynamic program
988	instances proved optimal by two methods
41,514	pairwise method comparisons, none disagreeing
16,920	canonical campaign outcomes audited

Formal scope and remaining conjectures: THEOREM_AUDIT.md.